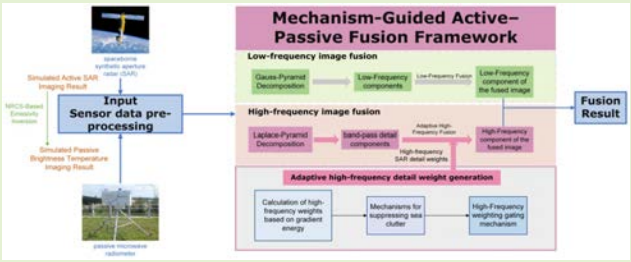
Multiscale Fusion of Heterogeneous SAR and Radiometer Data for Robust Maritime Target Enhancement

Yi-Dan Wang, Tan Qu, Yan Zhang, Jia-Ji Wu, *Member, IEEE*, and Zhen-Sen Wu, *Senior Member, IEEE*

***Abstract*— To overcome the limitations of single-modality imaging in maritime ship detection, this article proposes a multiscale active–passive fusion framework. We develop a Laplacian-pyramid-based model to synergistically exploit complementary scattering and radiation information from heterogeneous active and passive data. The framework incorporates a soft ROI mask and a sea-clutter suppression mechanism to enhance target saliency and suppress background interference. A multidimensional evaluation framework with Pareto-frontier analysis is employed to determine the optimal operating point. Experimental results demonstrate that the proposed method not only outperforms single-modality imaging and representative methods including discrete wavelet transform (DWT), Laplacian pyramid (LP), convolutional neural network (CNN), and Transformer in target enhancement, boundary fidelity, and background suppression, but it also shows clear advantages when compared with PCA-FLF from the literature on SAR and microwave data fusion. Furthermore, we apply the proposed fusion method to a downstream binary ship classification and detection task based on the YOLO26 network, further validating its effectiveness in real detection scenarios. Ablation studies also confirm the contribution of each module within the proposed framework. The proposed active–passive fusion framework and analytical methodology provide a robust theoretical foundation for maritime ship detection under complex environments and adverse sea conditions.**



***Index Terms*— Laplacian pyramid (LP), Pareto optimization, ROI gating, sea-clutter suppression, sensor data fusion.**

I. Introduction

ACCURATE detection of sea surface ship targets provides the basis for obtaining key information such as position, scale, contour, and motion state, and supports maritime scene perception, surveillance, and situation assessment [1]. Active SAR sensors provide high-resolution scattering and geometric information but are sensitive to clutter and lack material or thermal perception [2]. Passive sensors retrieve radiation-related material and temperature information with concealment advantages but have limited spatial resolution [3].

Although existing maritime ship detection methods have improved robustness under complex backgrounds, single-sensor imaging still faces inherent limitations. SAR sensors provide high-resolution structural and scattering observations, but their measurements are susceptible to speckle and sea clutter, which may obscure target boundaries or introduce pseudodetail artifacts. Passive microwave radiometers provide brightness temperature observations that reflect the radiometric contrast between ships and the sea background, whereas their relatively low spatial resolution weakens contour and edge representation. Therefore, multisensor fusion of active SAR and passive microwave radiometric observations provides a promising way to exploit complementary scattering and radiometric information, to compensate for the limitations of individual sensors, and to improve the reliability of maritime target perception.

Existing studies on SAR and passive sensor data fusion can generally be summarized from two perspectives. One focuses on microwave radiometer and SAR collaboration, which is more closely related to the imaging modality in this work

Received 29 July 2026; accepted 14 August 2026. Date of publication 24 August 2026; date of current version 2 October 2026. This work was supported by the National Natural Science Foundation of China under Grant 62071359. The associate editor coordinating the review of this article and approving it for publication was Prof. Liang-Bi Chen. *(Corresponding author: Tan Qu.)*

Yi-Dan Wang, Tan Qu, and Jia-Ji Wu are with the School of Electronic Engineering, Xidian University, Xi'an 710071, China (e-mail: ydwang_021116@stu.xidian.edu.cn; tqu@xidian.edu.cn; wujj@mail.xidian.edu.cn).

Yan Zhang is with the School of Electronic Engineering, Xi'an Shiyou University, Xi'an 710065, China (e-mail: yzhang@xsyu.edu.cn).

Zhen-Sen Wu is with the School of Physics, Xidian University, Xi'an 710071, China (e-mail: wuzhs@mail.xidian.edu.cn).

This article has supplementary downloadable material available at https://doi.org/10.1109/JSEN.2026.3724923, provided by the authors.

Digital Object Identifier 10.1109/JSEN.2026.3724923

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and mainly serves geophysical parameter retrieval. The other focuses on optical, infrared, and SAR image fusion for image enhancement, classification, and detection tasks, providing methodological References for multiscale image fusion.

Research on the fusion of microwave radiometer sensor data and SAR can be traced back to the concept of active–passive microwave synergistic retrieval. Das et al. [4] first proposed to combine radiometer retrieval results with active microwave scattering information for soil moisture estimation. Fluhrer et al. [5] estimated surface roughness parameters of bare soil using joint observations. Karvonen [6] combined passive microwave data with SAR for sea-ice concentration estimation. These studies demonstrate the complementary relationship between microwave radiometer and SAR observations. Microwave radiometers provide brightness temperature and radiometric information, whereas SAR captures scattering and structural features. Overall, existing active–passive microwave studies have mainly focused on geophysical parameter retrieval, such as soil moisture, surface roughness, and sea-ice concentration. In comparison, image-level fusion for maritime ship targets has a different emphasis: it requires not only the use of complementary observations, but also the joint consideration of target saliency enhancement, boundary preservation, and SAR clutter or speckle suppression. Therefore, extending active–passive microwave collaboration to fast image fusion and target enhancement remains worthy of further investigation.

Image-level fusion studies on optical, infrared, and SAR sensor data provide useful references for multisource-sensor data fusion. Traditional methods usually employ multiscale decomposition to preserve background trends at low-frequency components and integrate edges, textures, or salient structures at high-frequency components [7]. Deep learning methods, such as convolutional neural network (CNN) [8] and Transformers [9], further improve cross-modal feature alignment and task-driven fusion. However, these methods are mainly designed for optical, infrared, and SAR data. Their imaging mechanisms, spatial-resolution relationships, and noise characteristics differ from those of microwave radiometer-SAR fusion. In maritime scenes, SAR high-frequency components contain both target boundaries and sea clutter, whereas passive microwave images provide brightness temperature contrast but weak structural details. Therefore, directly applying general fusion methods may not sufficiently balance target enhancement, boundary preservation, and clutter suppression.

Nevertheless, image-level fusion of active SAR and passive microwave T_B (brightness-temperature) data remains challenging because synchronized and spatially aligned observations are difficult to obtain in maritime scenarios. Conventional methods often ignore the physical relationship and distinct roles of the two modalities, allowing strong SAR sea-clutter responses to appear as pseudodetails in fused images. As illustrated in Fig. 1, these limitations motivate a physics- and measurement-based framework for active–passive fusion.

This study proposes a mechanism-guided framework that establishes a physics-based mapping from SAR scattering responses to T_B distributions. SAR provides structural information, while the generated T_B image supplies radiometric contrast. A soft ROI mask, together with target-confidence and sea-clutter-risk regulation, controls SAR high-frequency injection to preserve vessel structures and suppress background contamination. Pareto-frontier analysis is further used to balance competing fusion objectives. The framework requires neither network training nor large-scale annotations and integrates the two modalities according to their complementary physical roles rather than simple pixel-level superposition.

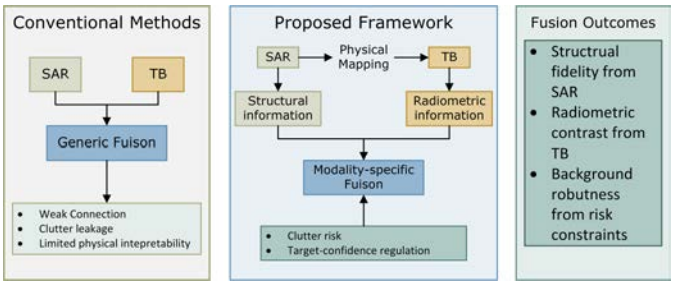


Fig. 1. Overview of the proposed active–passive fusion framework.

II. Methods

A. Active and Passive Imaging Simulation for Ship Targets

Due to external constraints such as satellite overpass schedules, weather conditions, and observation geometry, obtaining fully aligned, measured SAR and microwave radiometer data acquired over the same target at the same time and location is extremely challenging. To address this limitation, this study establishes a controllable, semiphysical simulation platform with accurate ground truth, enabling active and passive imaging simulations under consistent spatiotemporal conditions and providing a reliable basis for validating the effectiveness of the proposed fusion method.

1) Active SAR Simulation and Evaluation for Ship Targets: For active SAR imaging simulation, measured NRCS data of ships from the public OpenSARShip dataset [10] are used to construct the target scattering distribution. The NRCS map is converted into simulated SAR echoes according to the assumed observation geometry, where the echo amplitude is related to the square root of the backscattered power [11]. The range-Doppler, chirp scaling, and Omega-K algorithms are applied under different bandwidths and sampling rates, with key parameters set according to the OpenSARShip data and Sentinel-1 IW mode [12]. The specific simulation parameters are provided in Table S1 of the Supplementary Material.

Before evaluation, both the simulated and measured SAR images are normalized and converted into 8-bit grayscale images with pixel values ranging from 0 to 255. The simulated images are evaluated from two aspects: image quality, measured by mean, variance, equivalent number of looks (ENLs), and entropy, and similarity to measured SAR images, measured by the normalized mean-squared error (NMSE), peak signal-to-noise ratio (PSNR), structural similarity (SSIM), and cosine similarity. The evaluation results of simulated image quality and simulated–measured image similarity for different imaging schemes are provided in Tables S2 and S3 of the Supplementary Material. After spatial alignment,

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these metrics are integrated using a CRITIC-TOPSIS framework [13] to rank different imaging schemes and select the optimal one for subsequent active–passive fusion experiments. The weights of the evaluation indicators and the complete ranking of imaging schemes are provided in Tables S4 and S5 of the Supplementary Material.

2) Passive Imaging Simulation for Ship Targets: Passive imaging simulation is based on the target brightness temperature distribution, which depends on surface emissivity modeling. Surface emissivity is determined by roughness geometry parameters. When $\lambda/20 < \delta < \lambda/6$ [δ and λ denote root-mean square (rms) height of the tns of Pareto frontier and optimal operating points, target surface and wavelength], the surface is moderately rough, and the emissivity is calculated from the bistatic scattering coefficients based on energy conservation [14]

$$e_p(\theta) = 1 - \frac{1}{4\pi\cos\theta}\int_0^{\pi/2}\int_0^{2\pi} \times \left[\sigma_{pp}(\theta;\theta_s,\varphi_s) + \sigma_{pq}(\theta;\theta_s,\varphi_s)\right]\sin\theta_s\, d\theta_s\, d\varphi_s \quad (1)$$

where θ is the incidence angle. σ_{pp} and σ_{pq} are the scattering coefficients, where p and q represent horizontal or vertical polarization. θ_s and ϕ_s represent the elevation angle and azimuth angle of the scattering-field direction vector, respectively. When $\delta > \lambda/6$, the surface behaves as a Lambertian surface with diffuse scattering characteristics, and the emissivity is calculated [14]

$$e_p(\theta) = 1 - \frac{1}{4\pi\cos\theta}\int_0^{\pi/2}\int_0^{2\pi} \sigma_0^o \cos\theta\cos\theta_s\sin\theta_s\, d\theta_s\, d\varphi_s = 1 - \frac{\sigma_0^o}{4} \quad (2)$$

where σ_0^o is the scattering coefficient of a Lambertian surface.

According to sea wave statistics, the wave height is approximately four times the rms Height of the sea surface [15]. At the center frequency of 5.405 GHz [16], and under the selected Beaufort sea state, the estimated rms height is larger than this threshold, and the emissivity can therefore be calculated using the Lambertian model. For ship targets, the NRCS provides an observable scattering descriptor that reflects the electromagnetic response of each facet. Under a specified rough-surface scattering model, this descriptor is related to the equivalent roughness state of the facet. The equivalent rms height is then retrieved from the scattering model [17], followed by the calculation of the roughness-corrected emissivity.

Passive imaging is described using a radiometric transfer model. The surface-leaving brightness temperature ($T_{B_{\text{surf}}}$) consists of self-emission, reflected environmental radiation, and transmitted background radiation [18]

$$T_{B_{\text{surf}}} = e\cdot T + r\cdot T_s + t\cdot T_b \quad (3)$$

$$e + r + t = 1 \quad (4)$$

where T, T_s, and T_b denote the target physical temperature, surface-incident downwelling environmental brightness temperature, and effective upwelling atmospheric temperature, respectively. Considering atmospheric attenuation [19], the brightness temperature received by the sensor is

$$T_{B_{\text{obs}}} = \tau_a\cdot T_{B_{\text{surf}}} + (1-\tau_a)\cdot T_a \quad (5)$$

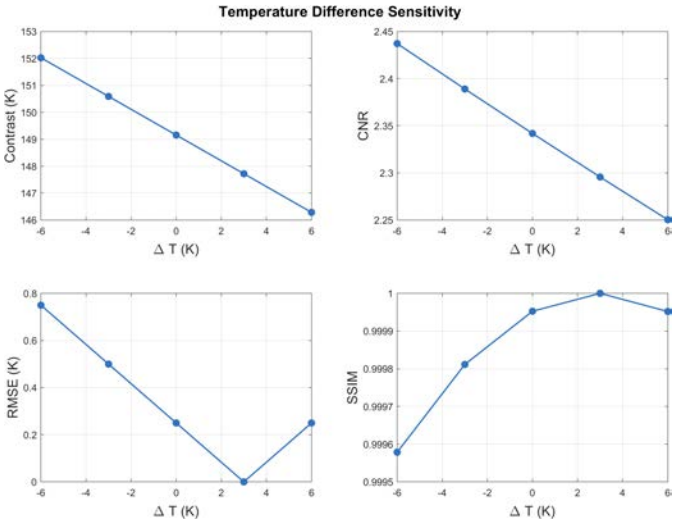


Fig. 2. Sensitivity to ΔT.

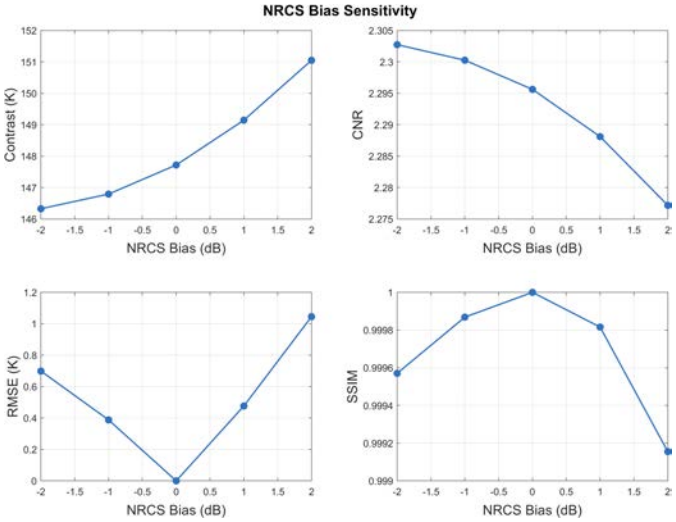


Fig. 3. Sensitivity to NRCS bias.

where τ_a and T_a denote atmospheric transmittance and the equivalent atmospheric temperature, respectively. For an opaque ship surface, $t = 0$ and $r = 1 - e$, giving

$$T_{B_{\text{obs}}} = \tau_a\left[e\cdot T + (1-e)\cdot T_s\right] + (1-\tau_a)\cdot T_a. \quad (6)$$

Sensitivity experiments were conducted for the ship-sea temperature difference ΔT, NRCS calibration bias, and atmospheric transmittance τ_a. As shown in Figs. 2–4 [20], [21], these perturbations mainly affect radiometric magnitude and vessel-sea brightness-temperature difference, while the spatial structure remains stable, with the SSIM exceeding 0.9948 in all tested cases. Atmospheric transmittance has the strongest influence on radiometric magnitude, ΔT has a relatively greater effect on CNR [21], and NRCS bias has only a limited effect on target-background separability. Because outlier suppression and robust normalization are applied before fusion, these predominantly radiometric variations are substantially reduced in the normalized inputs.

Accordingly, the simulation assumes that the ship surface is 3 °C warmer than the sea surface, sets $\tau_a = 1$, and retrieves

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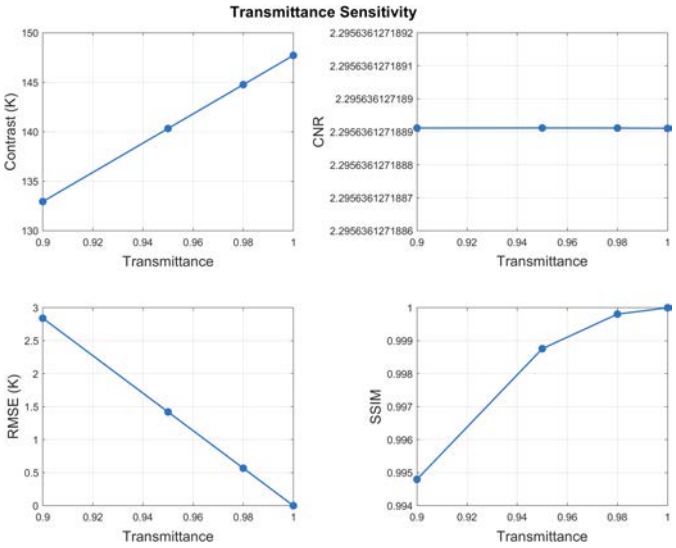


Fig. 4. Sensitivity to τ_a.

emissivity from NRCS. These assumptions provide controlled data for validating the fusion mechanism rather than a strict quantitative reconstruction of brightness temperature under all atmospheric conditions.

B. Fusion of Active and Passive Images for Sea Surface Ship Targets

The active and passive image fusion mechanism established in this article is illustrated in Fig. 5.

1) Physics-Based Active-Passive Input Preparation: Based on the physical consistency relationship between scattering and radiation, the emissivity is first retrieved from NRCS to obtain the brightness temperature image. The SAR amplitude has a large dynamic range and heavy-tailed distribution, which may cause extreme scatterers to dominate fusion. It is mapped to the logarithmic domain, and robust quantile clipping is applied to both the SAR amplitude and T_B to suppress outliers. Let the normalized SAR image in decibels and the T_B image be denoted as I_S and I_T, respectively. To align target bright regions, I_T is inverted.

2) Low Frequency Fusion Based on Gaussian Pyramid Decomposition: The Gaussian pyramid decomposes an image into progressively lower resolution levels using recursive low-pass filtering and downsampling [22]. Let I_S, I_T correspond to the original images G_0^S and G_0^T at level 0. After multiscale Gaussian pyramid decomposition, the corresponding low-frequency levels are obtained as

$$G_l^{\mathrm{S}} = \mathcal{D}_2\left(G_{l-1}^{\mathrm{S}} * \phi\right), G_l^T = \mathcal{D}_2\left(G_{l-1}^T * \phi\right) \quad l = 1, 2, \ldots, L. \tag{7}$$

$\mathcal{D}_2$ denotes the twofold downsampling operator, $*$ denotes 2-D convolution, and ϕ is the Gaussian kernel. The parameter L determines the multiscale coverage and is set to 4 here. After recursion, the low-frequency components of the two source images are obtained as G_L^S and G_L^T. Considering that the sea surface background tends to exhibit fluctuating textures at low frequency in SAR images, while the low frequency of T_B better reflects radiation distribution, the low-frequency fusion adopts a convex combination dominated by T_B

$$G_L^F = \alpha G_L^S + (1 - \alpha) G_L^T \tag{8}$$

where $\alpha \in [0, 1]$ controls the proportion of SAR contribution in the low-frequency fusion.

3) Adaptive Fusion of High-Frequency Details Based on LP Decomposition: In the high-frequency fusion stage, the Laplacian pyramid (LP) is employed to decompose the images, capturing multiscale detail information via bandpass residuals [23]. In the LP, the bandpass detail at level l is defined as the difference between the image at that level and its low-frequency prediction

$$H_l = G_l - \hat{G}. \tag{9}$$

$\hat{G}$ denotes the low-frequency prediction of the image at level l

$$\hat{G}_l^S = \mathcal{U}_2\left(G_{l+1}^S\right) * \phi, \quad \hat{G}_l^T = \mathcal{U}_2\left(G_{l+1}^T\right) * \phi \quad l = 0, 1, \ldots, L - 1 \tag{10}$$

where $\mathcal{U}_2$ denotes the twofold upsampling operator implemented by bilinear interpolation. The key implementation parameters of the proposed method are summarized in Table I.

TABLE I
KEY IMPLEMENTATION DETAILS OF THE PROPOSED METHOD

Implementation item	Parameter setting
Gaussian kernel	Four pyramid levels: $\sigma = 1.0$ pixel; kernel size 7×7
Quantile clipping	P1–P99 for SAR and T_B normalization; P99 for gradient-energy clipping
Sobel operator	Standard 3×3 Sobel operator
Computational complexity	Core fusion: $O(N)$ time and $O(N)$ memory; parameter search: $O(N_p N)$

To adaptively balance active SAR details and passive T_B details in high-frequency fusion, while suppressing sea-clutter and speckle noise, a high-frequency weighting mechanism is developed by incorporating local structural energy and an ROI (region of interest) prior gating strategy. The local gradient energy at pixel location (x, y) is defined as

$$E_l(x, y) = Avg_{\Omega_k}\left(\left(\partial_x |H_l|\right)^2 + \left(\partial_y |H_l|\right)^2\right) \tag{11}$$

where Ω_k denotes a local window of size $k \times k$ and Avg is the window mean. After robust compression, upper quantile clipping, and power flattening of the local gradient energy to obtain $\overline{E}$, a logarithmic energy ratio r is constructed, and a sigmoid mapping yields the initial SAR detail weight w, where τ is a temperature parameter controlling the sharpness of the weight transition, and ϵ is a stabilization term

$$r = log\left(\overline{E}_S + \epsilon\right) - log\left(\overline{E}_T + \epsilon\right) \tag{12}$$

$$w = 1/\left[1 + \exp\left(-\frac{r}{\tau}\right)\right]. \tag{13}$$

Due to strong high-frequency fluctuations of sea clutter that produce speckle, relying only on the energy ratio may cause SAR high-frequency responses from the background

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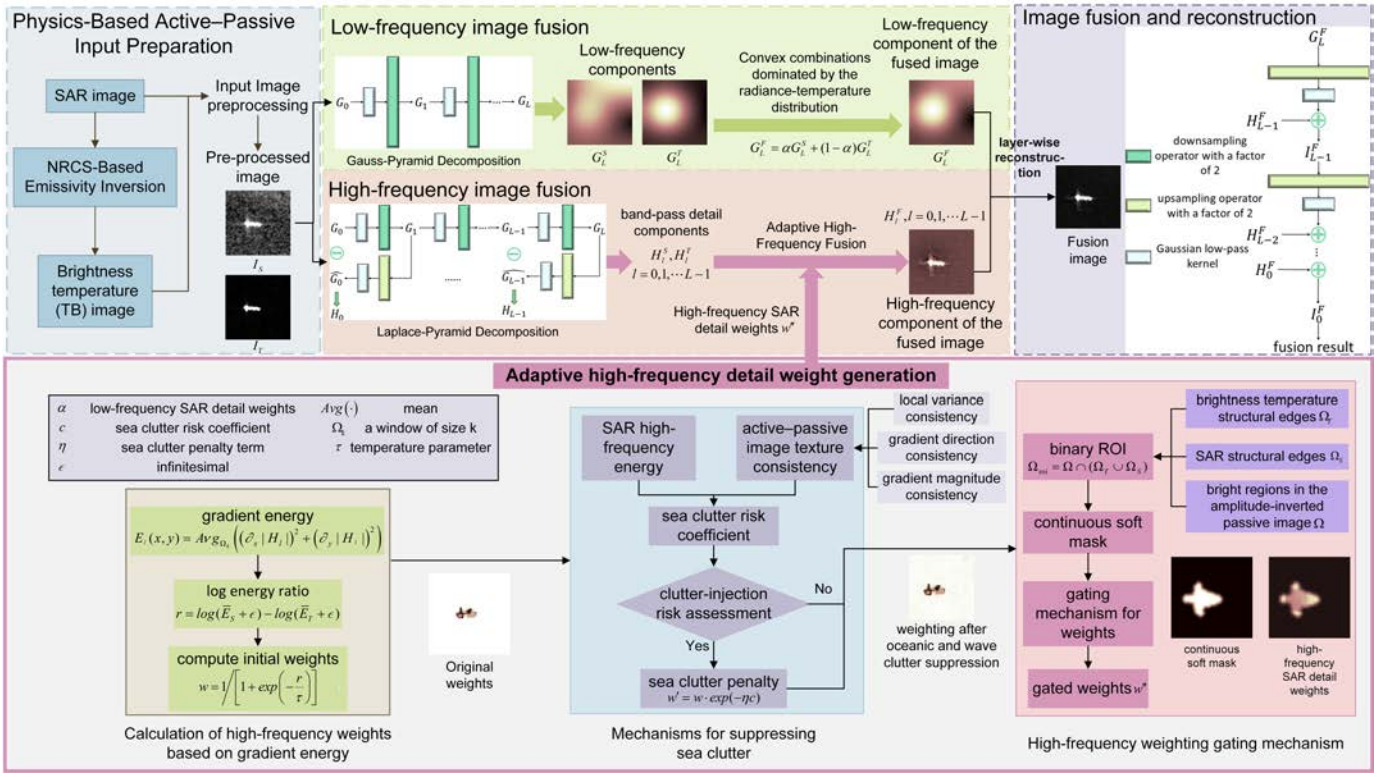


Fig. 5. Block diagram of the fusion mechanism.

to dominate fusion. A sea-clutter suppression mechanism is therefore introduced by constructing a risk coefficient based on local texture consistency and SAR high-frequency energy. Regions with high risk are penalized by applying an exponential penalty to the weights to suppress clutter-induced details

$$w' = w \cdot \exp(-\eta c), \eta > 0, c \in [0, 1] \tag{14}$$

where η denotes the penalty strength, and c represents the sea-clutter risk coefficient, which increases as texture consistency decreases and SAR high-frequency energy increases.

To extract spatial priors of target regions from complementary active and passive information and represent the confidence of foreground membership in a continuous form, an ROI mask is constructed to enable adaptive enhancement of target details and suppression of background clutter in subsequent fusion. High-intensity regions in the inverted image I_T are extracted as initial candidates Ω. A structural consistency gate extracts brightness temperature edges Ω_T using the Sobel gradient magnitude and obtains SAR edges Ω_S by the local averaging of Sobel gradient energy. The binary ROI is defined as

$$\Omega_{\text{roi}} = \Omega \cap (\Omega_T \cup \Omega_S). \tag{15}$$

To enable smooth gating between target and background and reduce boundary artifacts, the processed Ω_{roi} is Gaussian smoothed to obtain a soft mask $\rho(x)$. Using this mask, background and target weights are adaptively modulated based on pixelwise confidence, yielding the gated fusion weights

$$w''(x) = \left(1 - \rho(x)\right) w'_{\text{bg}}(x) + \rho(x)\, w'_{\text{fg}}(x). \tag{16}$$

Background regions use a compressed mapping to form the background constraint weight w'_{bg} to reduce SAR high-frequency contribution, while target regions use upper and lower bounds to form the foreground constraint weight w'_{fg} to preserve target details. Based on the gated weight w'', high-frequency fusion is performed at each scale

$$H_l^F = w'' H_l^S + \left(1 - w''\right) H_l^T. \tag{17}$$

4) Fusion Reconstruction: Fusion reconstruction starts from G_L^F and progressively upsamples while adding H_l^F from coarse to fine, recovering the original resolution and yielding the final fused image at the finest scale. Let the low-frequency base be the fused image at the coarsest layer, namely $I_L^F = G_L^F$

$$I_l^F = U_2\left(I_{l+1}^F\right) * \phi + H_l^F, \quad l = 0, 1, \ldots, L-1 \tag{18}$$

where ϕ is the same Gaussian kernel as defined above.

III. Results and Discussion

A. Adaptive Optimization Based on Pareto Frontier

Since the proposed fusion method involves multiple tunable parameters, different configurations affect low-frequency allocation, high-frequency enhancement, and background suppression. Therefore, parameter selection is not a single-objective problem, but a tradeoff among target enhancement, edge preservation, and background noise suppression. Pareto frontier analysis is employed to screen parameter combinations [24]. By varying the low-frequency fusion weight α, high-frequency energy compression exponent γ, weight temperature parameter τ, sea-clutter penalty strength η, and high-frequency weights for target and background regions, different fusion

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TABLE II
VESSEL INFORMATION

Acquisition time	Vessel name	Vessel type	Geographic coordinates
2015.09.16	HUA YUAN 11	Cargo	(119.167, 38.904)
2016.01.02	STELLAR LIBERTY	Cargo	(119.207, 38.904)
2016.02.19	ZHONGGANGTUO702	Tug	(118.306, 38.864)
2016.03.26	SABRINA	Tanker	(117.962, 38.914)
2016.04.05	KANG HING	Cargo	(114.145, 22.150)
2016.05.25	ANNAMARIA	Cargo	(118.225, 38.945)
2016.07.12	bei hai jiu 117	Search	(118.383, 38.924)
2016.08.17	YM ETERNITY	Tug	(118.630, 38.868)
2016.10.18	xinhaixing55	Cargo	(121.796, 28.722)

results are obtained, and Pareto optimal solutions are identified. The parameter set closest to the ideal point after metric normalization is selected, achieving a balanced tradeoff among target saliency, edge preservation, and background clutter suppression, and is used for active and passive image fusion.

1) Comprehensive Evaluation Metrics for Images: Starting from the core objectives of image fusion, this article evaluates performance from three aspects, namely target saliency, edge preservation, and background clutter suppression. To comprehensively reflect these aspects, five quantitative metrics are adopted, including the signal-to-clutter ratio (SCR) [25], contrast measure (CM) [26], edge strength (ES) [27], the background coefficient of variation (CV_b) [27], and ENLs in the background region (ENL_b) [26], which are used to select the optimal solution from the Pareto frontier.

2) Visualization of Pareto Frontier and Fusion Results: To comprehensively validate the effectiveness of the proposed method under diverse scenarios, the nine datasets listed in Table II are selected as study cases. The experiments cover different weather conditions, vessel types, and image sizes, supporting the assessment of method generalization. The meteorological conditions at the imaging time of each sea-surface ship target are shown in Table S6 of the Supplementary Material, and the corresponding emissivity distributions are shown in Fig. S1.

The Pareto frontiers of the fusion results for each ship target and the corresponding optimal parameter points are shown in Fig. 6. The selected operating point was determined according to its cross-case ranking consistency and local parameter stability, rather than the optimum of any single vessel case. Since CV_b is inversely related to background noise suppression performance, to maintain a consistent maximization criterion in the Pareto frontier analysis, $-CV_b$ is adopted as the corresponding optimization metric.

The normalized single-modality active and passive images and the corresponding optimal fusion results for each ship target are shown in Fig. 7 and Fig. S2. Compared with single-modality images, SAR preserves contours and scattering details but suffers from strong speckle and sea clutter, while T_B images provide smoother backgrounds and higher saliency but weaker edges and structures. The fused images achieve a better balance between target saliency and structural clarity.

TABLE III
QUANTITATIVE COMPARISON OF ACTIVE, PASSIVE, AND FUSED IMAGES FOR STELLAR LIBERTY

Image type	$SCR\uparrow$	$CM\uparrow$	$ES\uparrow$	$CV_b\downarrow$	$ENL_b\uparrow$
SAR	2.974	0.370	**0.606**	0.396	6.394
T_B	**14.521**	**0.830**	0.175	0.678	2.209
Proposed	13.099	0.674	0.337	**0.310**	**10.383**

B. Comprehensive Evaluation of Fused Images

To demonstrate the effectiveness of the proposed method in enhancing target saliency, preserving target edges, and suppressing background noise, evaluation experiments are designed to validate its performance advantages from multiple perspectives.

1) Comparison Experiments With Single Modality Images: The quantitative evaluation results of the fused images obtained by the proposed method and the corresponding single-modality active and passive images are presented in Table III. The comparison results for the remaining targets are provided in Tables S7–S14 of the Supplementary Material.

Compared with SAR images, the fused images improve saliency, with target–background separability increasing by 2–8 times and relative gray level difference by 1.5–2 times. Although edge responses decrease by 35%–55%, target boundaries remain identifiable. The background coefficient of variation decreases by 15%–60%, and the ENLs increase by 1.4–6 times, indicating improved smoothness and clutter suppression. Compared with T_B images, saliency decreases by 15%–30%, while contrast lies between the two. Boundary and structural information improve by 2 and 3 times, and the background coefficient of variation decreases by 65%–75% with the ENLs increasing by 9–14 times, resulting in a more uniform background. Overall, the fused images achieve a balanced tradeoff and outperform single-modality images.

2) Comparison With Classical Fusion Methods: To validate the effectiveness of the proposed fusion method, it is compared with representative methods including discrete wavelet transform (DWT) [28], LP [29], CNN [30], and Transformer [31].

For DWT, three-level db2 decomposition was adopted, with weighted averaging for low-frequency coefficients and local-energy-based weighting for high-frequency coefficients. LP used a four-level LP with a 5×5 Gaussian kernel ($\sigma = 1.0$) and local-energy-based detail fusion. The CNN comprised two modality-specific encoders with channel dimensions of 32, 64, and 128, followed by feature concatenation and convolutional reconstruction. The Transformer used 8×8 patches, 64-D embeddings, four blocks, four attention heads, an MLP ratio of 4, and cross-modal attention. Both learning-based baselines were trained using the same training, validation, and testing partitions, with ship cases separated at the scene level to prevent information leakage. The input images were spatially aligned, resized to the same resolution, and normalized using the same preprocessing procedure. The networks were trained from scratch using the Adam optimizer for 200 epochs, with a batch size of 8 and an initial learning rate of 0.0001. The

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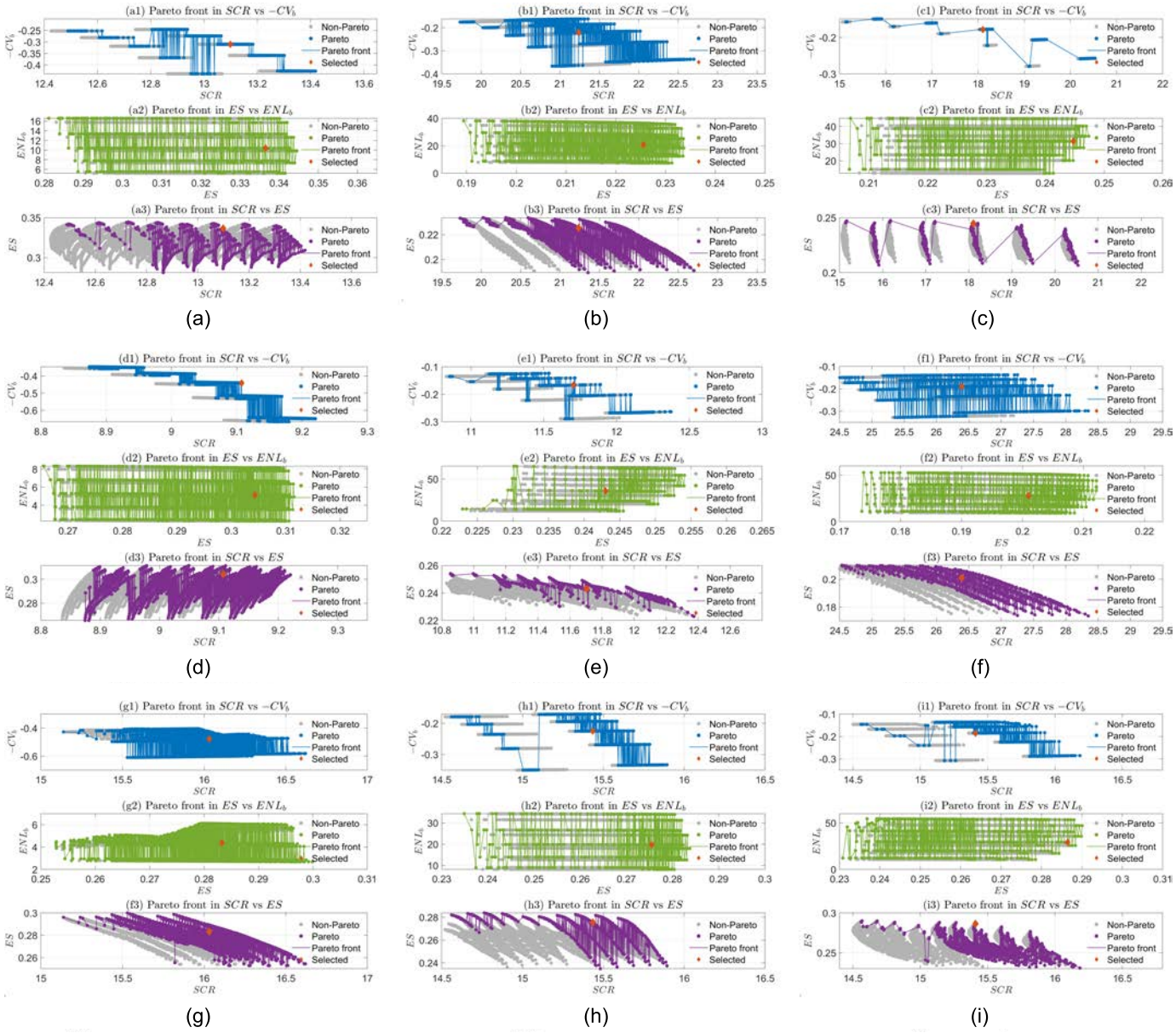


Fig. 6. Distributions of Pareto frontier and optimal operating points, (a) stellar liberty, (b) sabrina, (c) annamaria, (d) YM eternity, (e) xinhaixing55, (f) hua yuan 11, (g) zhonggangtuo702, (h) kang hing, and (i) bei hai jiu 117.

learning rate was reduced using a cosine-decay schedule. A combination of intensity, structural similarity, and gradient losses was adopted to preserve global radiometric information and local structural details. Model selection was performed according to the validation loss, and no test images were used for architecture selection or hyperparameter tuning.

The fusion results of different methods are shown in Fig. 8 and Fig. S3, and the quantitative evaluation results are presented in Table IV. The comparison results for the remaining targets are provided in Tables S15–S22 of the Supplementary Material. Analysis shows that the proposed method improves target background separability and contrast, enhances target prominence, and suppresses clutter and speckle while preserving structural information. It achieves a balance between saliency and background smoothness, demonstrating good effectiveness and robustness.

TABLE IV
QUANTITATIVE COMPARISON OF CLASSICAL FUSION METHODS FOR ANNA MARIA

Fusion method	$SCR\uparrow$	$CM\uparrow$	$ES\uparrow$	$CV_b\downarrow$	$ENL_b\uparrow$
Proposed	**19.190**	**0.693**	0.237	**0.207**	**23.348**
DWT	3.097	0.512	0.336	0.680	2.165
LP	2.305	0.329	**0.409**	0.428	5.463
CNN	3.602	0.495	0.333	0.545	3.370
Transformer	3.486	0.413	0.280	0.404	6.121

3) Comparison With a PCA-FLF Fusion: To provide a comparison with an existing multisensor fusion strategy, a PCA-based feature-level fusion baseline (PCA-FLF) was developed following the general idea of Khachatrian et al. [32]. SAR intensity, SAR texture features, and brightness-temperature

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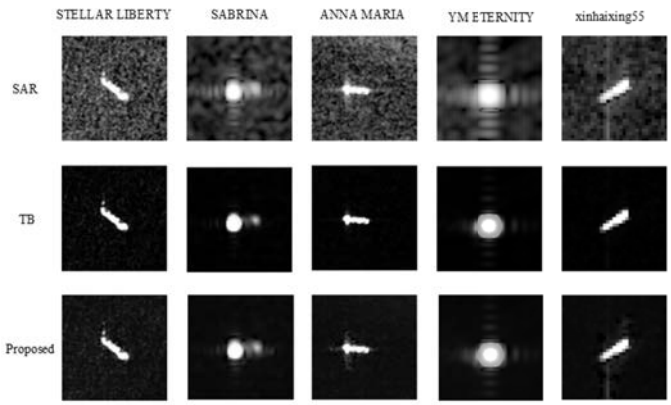


Fig. 7. Comparison between the proposed method, active, and passive images.

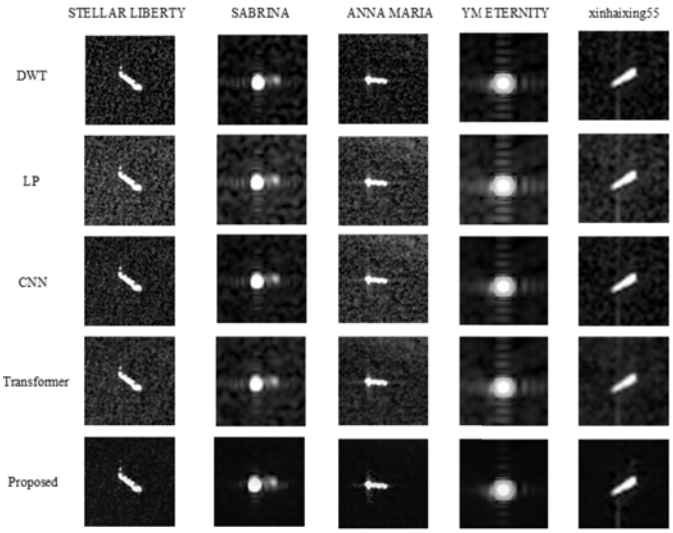


Fig. 8. Comparison of classical fusion methods.

TABLE V
QUANTITATIVE COMPARISON OF PCA-FLF FUSION METHOD FOR ANNA MARIA

Image types	$SCR\uparrow$	$CM\uparrow$	$ES\uparrow$	$CV_b\downarrow$	$ENL_b\uparrow$
Proposed	**19.19**	0.693	**0.237**	**0.207**	**23.348**
PCA-based feature fusion	17.566	**0.844**	0.077	0.609	2.698

data were combined and projected onto a single fused image using PCA. As the original study focused on classification, PCA-FLF is regarded as an adapted baseline rather than a direct reproduction of the original method. The quantitative evaluation results are presented in Table V and Tables S23–S30. Overall, the proposed method provides better structural preservation and background stability, whereas PCA-FLF achieves higher global contrast. These results suggest that the proposed method offers a more balanced fusion result by preserving vessel information while suppressing background fluctuations.

4) Ablation Study: To verify the effectiveness and design rationality of the proposed method and clarify the contribution of each component, ablation experiments are conducted under consistent conditions. The impact of different mechanisms on the tradeoff among target saliency, background noise suppression, and boundary fidelity is analyzed. The averaged ablation results are summarized in Table VI, and a visual comparison of the proposed method and its ablated variants is presented in Fig. 9.

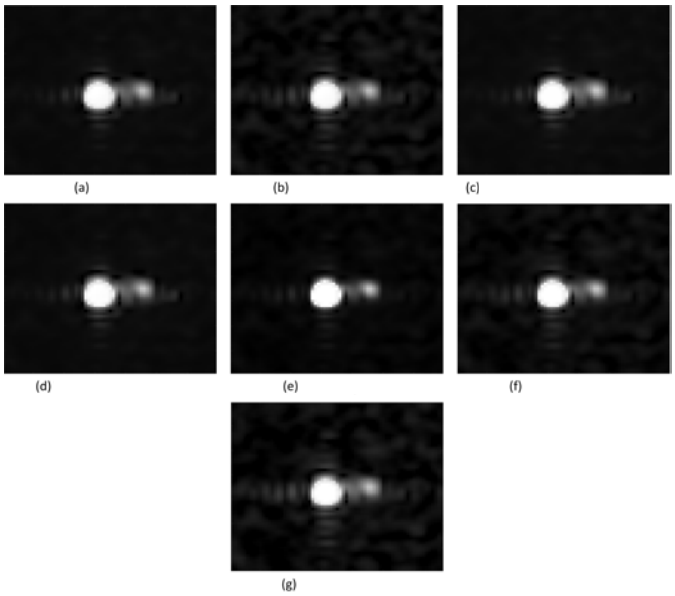


Fig. 9. Visual comparison of the proposed method and its ablated variants for sabrina: (a) complete proposed method, (b) without ROI gating, (c) without sea-clutter suppression, (d) without energy-based robustness, and (e)–(g) fixed-weight configurations.

TABLE VI
AVERAGE ABLATION STUDY RESULTS

Fusion method	$SCR\uparrow$	$CM\uparrow$	$ES\uparrow$	$CV_b\downarrow$	$ENL_b\uparrow$
Proposed	**14.867**	0.708	0.269	**0.269**	**19.099**
Without ROI gating	6.183	0.704	**0.317**	0.758	1.793
Without sea clutter suppression	14.812	0.708	0.275	0.269	18.994
Without energy-based robustness	18.840	**0.709**	0.274	0.270	19.083
Fixed weight 0.1	11.410	0.764	0.154	0.526	3.938
Fixed weight 0.3	7.343	0.704	0.230	0.631	2.637
Fixed weight 0.5	5.746	0.699	0.307	0.802	1.623

The comparison results for the remaining targets are provided in Tables S31–S35 of the Supplementary Material. Overall, the proposed method achieves a stable balance between target saliency and background suppression. Removing ROI gating causes the most pronounced degradation in target saliency and background homogeneity, demonstrating its dominant role in differentiated target–background processing. Fixed weighting provides a less favorable balance between target enhancement and background suppression, supporting the use of adaptive, ROI-gated multiscale weighting.

By comparison, sea-clutter suppression and energy-based robustness have limited and nonuniform effects on the averaged global metrics, mainly serving as auxiliary constraints on adaptive weight estimation. They suppress unsupported, high-energy SAR responses and reduce sensitivity to isolated strong scatterers. Their effects are less evident in smooth scenes or with reliable ROI gating. Under challenging conditions, such

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TABLE VII
DISTRIBUTION OF THE SHIP DETECTION DATASET

Category	Training	Validation	Test
Total	643	145	85
Cargo/Non-Cargo	406/237	120/25	58/27

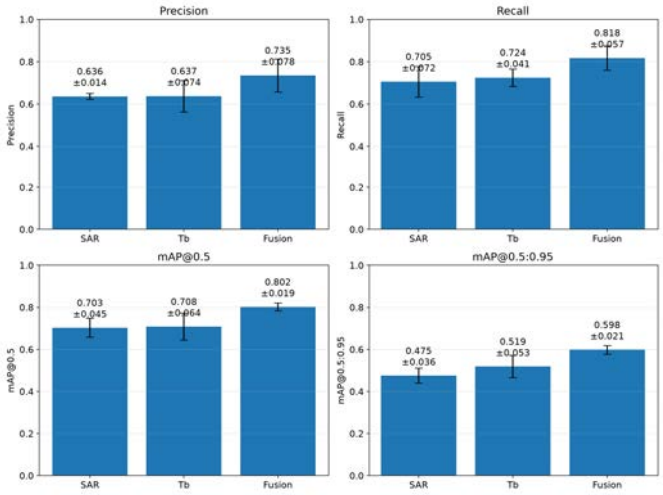


Fig. 10. Comparison of cargo and noncargo ship detection performance using SAR, T_B, and fused images. Error bars represent the standard deviations over five random seeds.

as strong speckle and abrupt clutter peaks, these modules are designed to reduce unsupported SAR detail injection and limit the influence of isolated strong scatterers on weight estimation. Together, they reduce clutter leakage and abnormal weighting in challenging scenes. Although removing either component may slightly improve some contrast- or edge-related metrics by retaining more SAR details, it also increases sensitivity to clutter peaks and cross-modal inconsistencies.

5) Downstream Validation on Cargo and Noncargo Ship Detection: To evaluate the practical value of the proposed method for downstream tasks, we constructed a two-class ship-detection dataset (Cargo and Noncargo) based on the OpenSARShip dataset. A reliable evaluation demands a sufficiently large number of samples, yet high-fidelity SAR simulation for many ship targets is computationally expensive. We therefore adopted measured SAR images as the active-modal inputs. From these measured SAR scenes, we generated the corresponding simulated T_B images and then obtained the fused images using the proposed method. Table VII shows the dataset distribution. We separately used the SAR, T_B, and fused images to train and test the YOLO26 network under identical dataset partitions and training configurations. Table VIII summarizes the network training parameters. To reduce the influence of randomness, we employed five random seeds for each input type and report the results as the mean and standard deviation.

As shown in Fig. 10, the fused images consistently outperform the SAR and T_B inputs across all evaluation metrics. The results indicate that the proposed method effectively integrates complementary scattering and radiometric information [33], thereby improving the detection and classification of cargo and noncargo ships. This experimental design also provides a preliminary assessment of the applicability of the proposed method to measured maritime scenes.

TABLE VIII
TRAINING PARAMETERS OF THE YOLO26 NETWORK

Training parameter	Setting
Initial weights	yolo26n.pt
Input image size	640 × 640 pixels
Number of epochs	100
Batch size	16
Weight-saving interval	10 epochs

IV. CONCLUSION AND FUTURE WORK

This article presents a multiscale active–passive fusion framework for maritime ship detection. We develop a Laplacian-pyramid-based architecture with ROI gating and sea-clutter suppression to effectively exploit complementary scattering and radiation information. A multidimensional evaluation framework with Pareto-frontier analysis is used to determine the optimal operating point. Experimental results demonstrate that the proposed method improves target saliency, boundary fidelity, and background suppression compared with single-modality images and representative methods including DWT, LP, CNN, Transformer, and PCA-FLF-based approaches. Ablation studies validate the effectiveness of the proposed components. Rather than simply performing pixel-level superposition, the proposed method exploits the deep complementarity between the physical sensing mechanisms of active SAR scattering and passive radiometric emission, providing an effective active–passive sensor data fusion strategy for maritime monitoring and robust target perception in complex environments.

To further examine the practical value of the proposed fusion framework, a downstream cargo and noncargo ship detection experiment was conducted using the YOLO26 network. The detector was separately trained and evaluated with measured SAR images, simulated T_B images, and the corresponding fused images. The results show that the fused inputs generally achieved better detection performance than the single-modal inputs under the current experimental settings. This finding provides preliminary evidence that the proposed fusion method can enhance target-related information and may benefit downstream maritime ship detection tasks. Nevertheless, the present validation is still limited to the constructed dataset and experimental configuration, and further evaluation using larger-scale, fully synchronized real active–passive observations is required to confirm its generalization capability.

Although the proposed framework achieves promising performance, its effectiveness may decrease under extreme environmental conditions. In severe sea states, strong and spatially nonstationary sea clutter may exceed the suppression capability of the current weighting strategy, allowing residual clutter to enter the fused image. In addition, when the thermal

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or emissivity contrast between the ship and the sea surface is weak, the passive T_B image may no longer provide a reliable target prior, thereby reducing the accuracy of ROI construction and weakening target enhancement. Future work will therefore investigate more adaptive clutter modeling and uncertainty-aware passive-prior estimation for complex maritime environments.

The current framework also assumes that the SAR and T_B images are accurately registered before fusion. Noticeable cross-modal misalignment may disrupt the structural correspondence used for ROI localization and adaptive weight estimation, resulting in shifted target regions, blurred boundaries, ghost artifacts, and degraded target–background separability. Future research will focus on robust multimodal registration and alignment-confidence-aware fusion, so that the contribution of cross-modal information can be adaptively reduced when the registration reliability is low.

REFERENCES

[1] W.-J. Lee et al., "Detection and tracking for the awareness of surroundings of a ship based on deep learning," *J. Comput. Design Eng.*, vol. 8, no. 5, pp. 1407–1430, Sep. 2021.

[2] K. Huang, J. Ding, and W. Deng, "An overview of millimeter-wave radar modeling methods for autonomous driving simulation applications," *Sensors*, vol. 24, no. 11, p. 3310, May 2024.

[3] Y. Meng, A. Qing, C. Lin, Z. Jiefeng, Y. Zhao, and C. Zhang, "Passive millimeter wave imaging system based on helical scanning," *Sci. Rep.*, vol. 8, no. 1, May 2018, Art. no. 7852, doi: 10.1038/s41598-018-25637-9.

[4] N. N. Das, D. Entekhabi, and E. G. Njoku, "An algorithm for merging SMAP radiometer and radar data for high-resolution soil-moisture retrieval," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 5, pp. 1504–1512, May 2011.

[5] A. Fluhrer, T. Jagdhuber, R. Akbar, P. E. O'Neill, and D. Entekhabi, "Simultaneous retrieval of surface roughness parameters for bare soils from combined active–passive microwave SMAP observations," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8182–8194, Oct. 2021.

[6] J. Karvonen, "Baltic sea ice concentration estimation using Sentinel-1 SAR and AMSR2 microwave radiometer data," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2871–2883, May 2017.

[7] X. Gong, Z. Hou, Y. Wan, Y. Zhong, M. Zhang, and K. Lv, "Multispectral and SAR image fusion for multiscale decomposition based on least squares optimization rolling guidance filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5401920.

[8] R. Liu, J. Ling, and H. Zhang, "SoftFormer: SAR-optical fusion transformer for urban land use and land cover classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 218, pp. 277–293, Dec. 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271624003502

[9] J. Li et al., "Fusion of optical and SAR images based on deep learning to reconstruct vegetation NDVI time series in cloud-prone regions," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 112, Aug. 2022, Art. no. 102818. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1569843222000206

[10] L. Huang et al., "OpenSARShip: A dataset dedicated to Sentinel-1 ship interpretation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 1, pp. 195–208, Jan. 2018.

[11] S. Wagner, M. A. Hossain, S. Pancrazio, and A. Pham, "Systembased specifications for better design of groundpenetrating radar antennas," *IET Microw., Antennas Propag.*, vol. 17, no. 6, pp. 478–493, May 2023, doi: 10.1049/mia2.12358.

[12] A. Schubert, N. Miranda, D. Geudtner, and D. Small, "Sentinel-1A/B combined product geolocation accuracy," *Remote Sens.*, vol. 9, no. 6, Jun. 2017, Art. no. 607, doi: 10.3390/rs9060607.

[13] J. Yin, J. Wang, C. Wang, L. Wang, and Z. Chang, "CRITIC-TOPSIS based evaluation of smart community governance: A case study in China," *Sustainability*, vol. 15, no. 3, p. 1923, Jan. 2023.

[14] F. T. Ulaby, R. K. Moore, and A. K. Fung, *Microwave Remote Sensing: Active and Passive, Vol. II: Radar Remote Sensing and Surface Scattering and Emission Theory*. Norwood, MA, USA: Artech House, 1982.

[15] K. Bärfuss, B. Djath, A. Lampert, and J. Schulz-Stellenfleth, "Airborne LiDAR measurements of sea surface properties in the German bight," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 4608–4617, Jun. 2021.

[16] H. Johnsen, R. Piantanida, and G. Hajduch, "Sentinel-1 product specification for products generated with IPF 3.4.0," CLS S-1 Mission Performance Center, Brest, France, Tech. Rep. S1-RS-MDA-52-7441-3-9/MPC-0240, ver. 3.9.1, May 2021.

[17] X. Zheng et al., "Simultaneously estimating surface soil moisture and roughness of bare soils by combining optical and radar data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 100, Aug. 2021, Art. no. 102345.

[18] N. Risse, M. Mech, C. Prigent, G. Spreen, and S. Crewell, "Assessing sea ice microwave emissivity up to submillimeter waves from airborne and satellite observations," *Cryosphere*, vol. 18, no. 9, pp. 4137–4163, Sep. 2024, doi: 10.5194/tc-18-4137-2024.

[19] F. J. Wentz and T. Meissner, "Atmospheric absorption model for dry air and water vapor at microwave frequencies below 100 GHz derived from spaceborne radiometer observations," *Radio Sci.*, vol. 51, no. 5, pp. 381–391, May 2016.

[20] T. Letcher, C. Vuyovich, A. Langlois, and A. Roy, "Understanding uncertainty of snow radiative transfer modeling within a mixed deciduous and evergreen forest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8225–8235, 2021.

[21] S. Kundu, M. K. Bhuyan, K. F. MacDorman, N. K. Sharma, and M. Bhatt, "Despeckling of synthetic aperture radar images using linear-angular attention transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–11, 2025.

[22] H. Chen, X. Du, H. Huang, and T. Zhao, "A real-time high-resolution multi-focus image fusion algorithm based on multi-scale feature aggregation," *Appl. Sci.*, vol. 15, no. 13, Jun. 2025, Art. no. 6967, doi: 10.3390/app15136967.

[23] B. Qi et al., "Infrared and visible image fusion via sparse representation and adaptive dual-channel PCNN model based on co-occurrence analysis shearlet transform," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–15, 2025.

[24] X. Wu et al., "Efficiently approximating the Pareto frontier: Hydropower dam placement in the Amazon Basin," in *Proc. AAAI Conf. Artif. Intell.*, vol. 32, Feb. 2018, pp. 849–859, doi: 10.1609/aaai.v32i1.11347.

[25] S. Kim and J. Lee, "Scale invariant small target detection by optimizing signal-to-clutter ratio in heterogeneous background for infrared search and track," *Pattern Recognit.*, vol. 45, no. 1, pp. 393–406, Jan. 2012.

[26] S. N. Anfinsen, A. P. Doulgeris, and T. Eltoft, "Estimation of the equivalent number of looks in polarimetric synthetic aperture radar imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 11, pp. 3795–3809, Nov. 2009.

[27] W. Xue, L. Zhang, X. Mou, and A. C. Bovik, "Gradient magnitude similarity deviation: A highly efficient perceptual image quality index," *IEEE Trans. Image Process.*, vol. 23, no. 2, pp. 684–695, Feb. 2014.

[28] Y. Zhang and G. Hong, "An ihs and wavelet integrated approach to improve pan-sharpening visual quality of natural colour ikonos and quickbird images," *Inf. Fusion*, vol. 6, no. 3, pp. 225–234, Sep. 2005, doi: 10.1016/j.inffus.2004.06.009.

[29] J. Du, W. Li, B. Xiao, and Q. Nawaz, "Union Laplacian pyramid with multiple features for medical image fusion," *Neurocomputing*, vol. 194, pp. 326–339, Jun. 2016.

[30] Y. Zhang, Y. Liu, P. Sun, H. Yan, X. Zhao, and L. Zhang, "IFCNN: A general image fusion framework based on convolutional neural network," *Inf. Fusion*, vol. 54, pp. 99–118, Feb. 2019.

[31] J. Ma, L. Tang, F. Fan, J. Huang, X. Mei, and Y. Ma, "SwinFusion: Cross-domain long-range learning for general image fusion via Swin transformer," *IEEE/CAA J. Autom. Sinica*, vol. 9, no. 7, pp. 1200–1217, Jul. 2022.

[32] E. Khachatrian et al., "SAR and passive microwave fusion scheme: A test case on Sentinel-1/AMSR-2 for sea ice classification," *Geophys. Res. Lett.*, vol. 50, no. 4, p. 2022, Feb. 2023, doi: 10.1029/2022gl102083.

[33] Y. Han et al., "Multi-ship detection and classification with feature enhancement and lightweight fusion," *Sci. Rep.*, vol. 15, Oct. 2025, Art. no. 38075, doi: 10.1038/s41598-025-21887-6.

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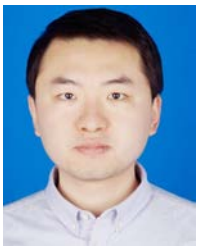
Yi-Dan Wang received the B.S. degree from Xidian University, Xi'an, China, in 2024, where she is currently pursuing the M.S. degree with the School of Electronic Engineering.

Her research interests include radar imaging of targets and its applications.



Tan Qu received the B.S. degree in electronic information science and technology and the Ph.D. degree in radio physics from Xidian University, Xi'an, China, in 2011 and 2016, respectively.

In 2016, she joined the School of Electronic Engineering, Xidian University. She is now an Associate Professor at Xidian University. Her research interests include electromagnetic wave propagation and scattering, computational electromagnetics, and remote sensing.



Yan Zhang received the B.S. degree in detection guidance and control technology and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 2010 and 2017, respectively.

In 2017, he joined the School of Electronic Engineering, Xi'an Shiyou University, Xi'an. His research interests include SAR imaging, electromagnetic scattering, and motion compensation.



Jia-Ji Wu (Member, IEEE) received the B.S. degree in electrical engineering from Xidian University, Xi'an, China, in 1996, the M.S. degree in astrometry and celestial mechanics from the National Time Service Center, Chinese Academy of Sciences, Xi'an, in 2002, and the Ph.D. degree in signal and information processing from Xidian University, in 2005.

He is currently a Professor with Xidian University. His research interests include still image coding, hyperspectral/multispectral image processing, communication, big data, the Internet of Things (IoT), and high-performance computing.



Zhen-Sen Wu (Senior Member, IEEE) received the B.S. degree in applied physics from Xi'an Jiaotong University, Xi'an, China, in 1969, and the M.S. degree in space physics from Wuhan University, Wuhan, China, in 1981.

From 1995 to 2001, he was a Visiting Professor with Rouen University, Rouen, France, for implementing joint study of two projects supported by the Sino-France Program for Advanced Research. He is currently a Professor with Xidian University, Xi'an. His current research interests include electromagnetic and optical wave scattering and propagation in random media, computational electromagnetism, target and environment photoelectric characteristics, scattering properties from complex land-oceanic surface and remote sensing, electromagnetic wave propagation and modeling and prediction in troposphere and ionosphere, and high-performance computing.

Prof. Wu is a fellow of the Chinese Institute of Electronics, Beijing, China.

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